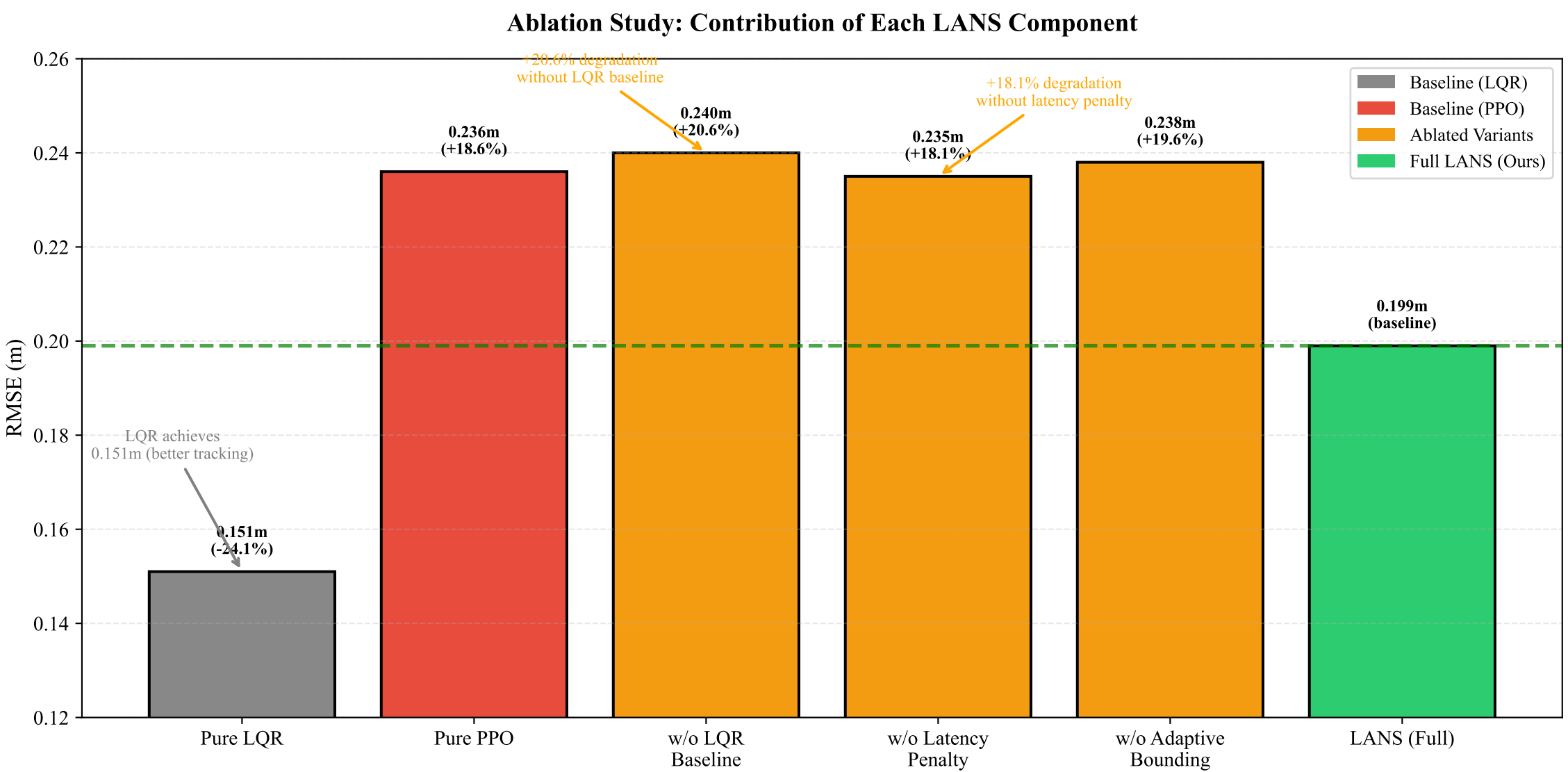
Ablation Study: Contribution of Each LANS Component
RMSE (m)
0.26
0.24
0.22
0.20
0.18
0.16
0.14
0.12
Baseline (LQR)
Baseline (PPO)
Ablated Variants
Full LANS (Ours)
LQR achieves
0.151m (better tracking)
0.151m
(-24.1%)
0.236m
(+18.6%)
+20.6% degradation
without LQR baseline
0.240m
(+20.6%)
+18.1% degradation
without latency penalty
0.235m
(+18.1%)
0.238m
(+19.6%)
0.199m
(baseline)
Pure LQR
Pure PPO
w/o LQR
Baseline
w/o Latency
Penalty
w/o Adaptive
Bounding
LANS (Full)